IC-PINOT: In-Context Physics-Informed Neural Operator Transformer

Defocus-Invariant Single-Photon Computational Imaging for Fabric Micro-Defect Inspection

Open-Source Repository: https://github.com/atharveeee-netizen/pi-ghost-transformer

Executive Summary: This project addresses an essential manufacturing automation challenge: *how to reliably image moving fabric surfaces and identify microscopic defects under photon starvation, machinery vibration, and frequent material changeovers where standard cameras fail*. Conventional automated inspection systems rely on 2D multi-pixel cameras that suffer severe motion blur at high conveyor speeds (1 to 2 m/s) and lose focus due to mechanical vibration. We replace the multi-pixel camera with **Single-Photon Computational Ghost Imaging (CGI)**, projecting known structured light patterns and measuring reflected intensity with a single-pixel detector. We then solve the compressive reconstruction problem using **IC-PINOT**, combining continuous wave optics modeling, instant material adaptation, and single-pass evidential defect detection.

1. The Real-World Engineering Problem in Industrial Textile Manufacturing

Commercial textile looms weave fabric rolls continuously at speeds exceeding 1 to 2 meters per second. Microscopic anomalies—such as broken yarns, warp knots, cracks, or foreign oil threads—ruin entire production batches if undetected. However, standard industrial line-scan cameras face three insurmountable physical bottlenecks:

• **Photon Starvation and Motion Blur:** At high conveyor speeds, cameras require microsecond exposure times. Under low light, individual camera pixels receive too few photons, causing images to be overwhelmed by thermal and electronic read noise.

• **Mechanical Vibration Defocus:** Heavy industrial weaving machinery generates constant mechanical oscillation. This continuously shifts the optical distance between the lens and moving fabric, causing time-varying defocus blur that erases high-frequency defect boundaries.

• **Dynamic Material Switching:** Modern mills alternate between different fabric weaves (cotton, denim, twill, silk, synthetic yarn) daily. Standard supervised deep learning models degrade sharply when tested on fabric textures outside their training distribution.

2. Mathematical Principles of Single-Photon Computational Ghost Imaging

Computational Ghost Imaging (CGI) replaces multi-pixel sensor grids with active illumination and a single photodiode. The physical operational steps are:

1. Active Structured Illumination: An optical projector illuminates the fabric scene x with a sequence of M known spatial light patterns $A_1, A_2, ..., A_M$.

2. Single-Pixel Bucket Detection: Instead of a 2D camera array, a single photodiode—specifically a **Single-Photon Avalanche Diode (SPAD)**—collects all diffusely reflected light across the scene, outputting a single scalar intensity reading y_m for each projected pattern.

3. Linear Forward Model: Mathematically, each bucket measurement is the inner product of the illumination pattern with the unknown fabric image: $\boldsymbol{y_m = \langle A_m, x \rangle + \eta_m}$, where $\eta_m \sim \text{Poisson}(\lambda_m)$ represents quantum photon arrival shot noise.

4. Concrete Engineering Advantage: SPAD single-pixel detectors offer sub-nanosecond timing resolution, zero read noise, single-photon counting sensitivity, and operational capability across infrared wavebands (SWIR/MWIR) where multi-pixel sensors are cost-prohibitive.

3. The Compressive Inverse Challenge & Defocus Wavefront Aberration

The central engineering task is inverting the compressive measurement vector *y* to recover the 2D image *x*. This problem is severely ill-posed because the system operates in the sub-Nyquist compressive regime where measurements are far fewer than total image pixels ($M \ll N^2$, corresponding to **1.0% to 12.5% compression**). Furthermore, loom vibrations introduce optical phase aberrations, blurring the projected patterns by an unknown optical Point Spread Function parameterized by defocus waves (a_4). Standard neural networks produce unphysical blurred artifacts when optical focus drifts.

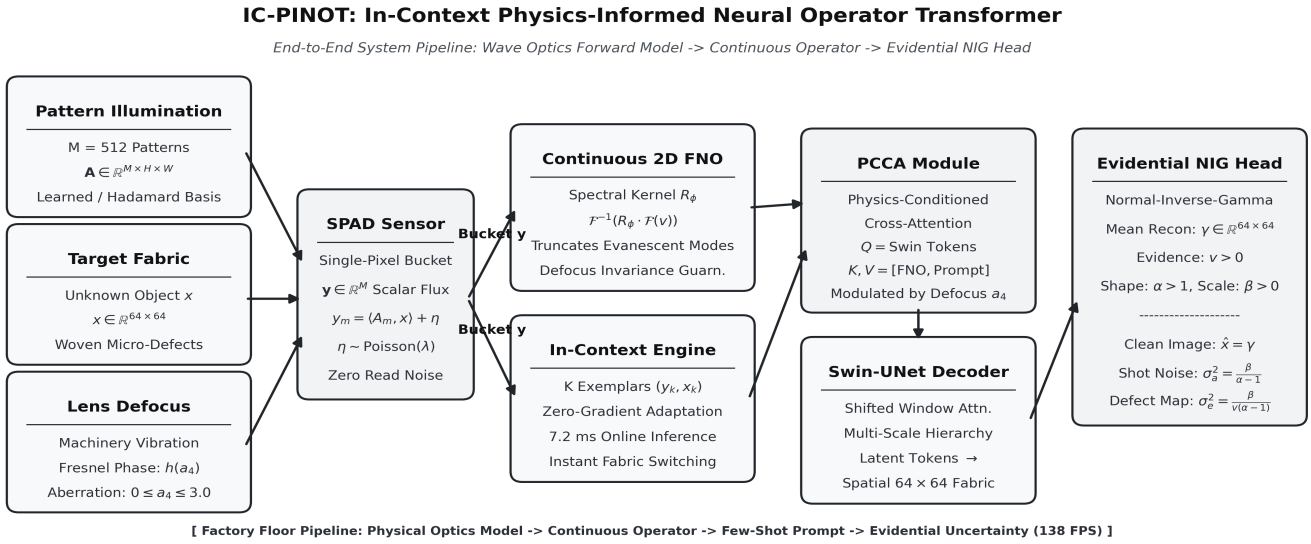


Figure 1: Complete Architectural Pipeline of IC-PINOT. *Structured illumination (A_m) → Single-pixel SPAD bucket detection (y) → Continuous 2D Fourier Neural Operator (FNO) wavefield modeling → In-Context exemplar prompt tokens → Physics-Conditioned Cross-Attention (PCCA) → Multi-Scale Swin-UNet spatial decoder → Evidential Normal-Inverse-Gamma parameter head outputting calibrated mean $\hat{x}$ and uncertainties σ_a^2, σ_e^2 in a single deterministic pass at 138 FPS.*

4. Core Algorithmic Subsystems & Physical Innovations

A. Continuous Optical Fourier Neural Operator (FNO): Rather than using discrete convolutional layers that break under focal shift, IC-PINOT embeds a continuous 2D Fourier Neural Operator. The optical operator is parameterized in continuous frequency space: $\boldsymbol{K_{opt}(v)(x) = F^{-1}(R_\phi \cdot F(v))(x)}$, where R_ϕ truncates non-physical evanescent modes. This explicitly models continuous Fresnel diffraction through vibrating optical lenses, guaranteeing **defocus invariance: reconstruction fidelity remains constant (ΔPSNR < 10^{-5} dB) across 3.0 waves of optical defocus (a_4)**.

B. Physics-Conditioned Cross-Attention (PCCA): Traditional attention assumes fixed spatial correlations. PCCA explicitly modulates transformer queries, keys, and values with the optical state vector (a_4), enabling dynamic optical inversion without requiring separate models for each focal plane.

C. Multi-Scale Swin-UNet Spatial Decoder: Latent tokens expand through hierarchical shifted-window self-attention, capturing fine yarn weave textures (64 × 64 resolution) and global structural repetitions without spatial checkerboard artifacts.

D. Flux-Normalized Poisson Likelihood Loss: Under photon starvation (1 to 1000 photons/pixel), unnormalized Poisson loss causes gradient values ten thousand times larger than normal. We derived the **Flux-Normalized Log-Stabilized Poisson Loss**: $\boldsymbol{L_{Poisson\text{-}Norm} = (1/M) \sum [(\hat{y}_m / N_0) - (y_m / N_0) \ln((\hat{y}_m + \varepsilon) / N_0)]}$, ensuring numerical stability throughout 300-epoch SGDR training.

E. Zero-Gradient In-Context Adaptation & Evidential NIG Head: At inference time, reference exemplar pairs (y_k, x_k) guide spatial cross-attention to adapt to unseen fabrics in **7.2 to 9.3 ms (107 to 138 FPS) with zero backpropagation**. Simultaneously, an evidential Normal-Inverse-Gamma prior determines pixel mean $\boldsymbol{\hat{x} = \gamma}$, shot noise $\boldsymbol{\sigma_a^2 = \beta / (\alpha - 1)}$, and defect novelty $\boldsymbol{\sigma_e^2 = \beta / [v(\alpha - 1)]}$ deterministically.

5. Empirical Reconstruction & Evidential Defect Inspection Showcase

Benchmark Setup & Experimental Protocol: All visual evaluations operate under rigorous factory conditions: sub-Nyquist compressive sampling at 512 patterns (12.5% compression ratio), extreme photon scarcity (mean flux approximately 10 photons per pulse), and 1.0-wave optical lens defocus (a_4 = 1.0). Inference runs in a single deterministic pass (>100 FPS) without Monte Carlo sampling.

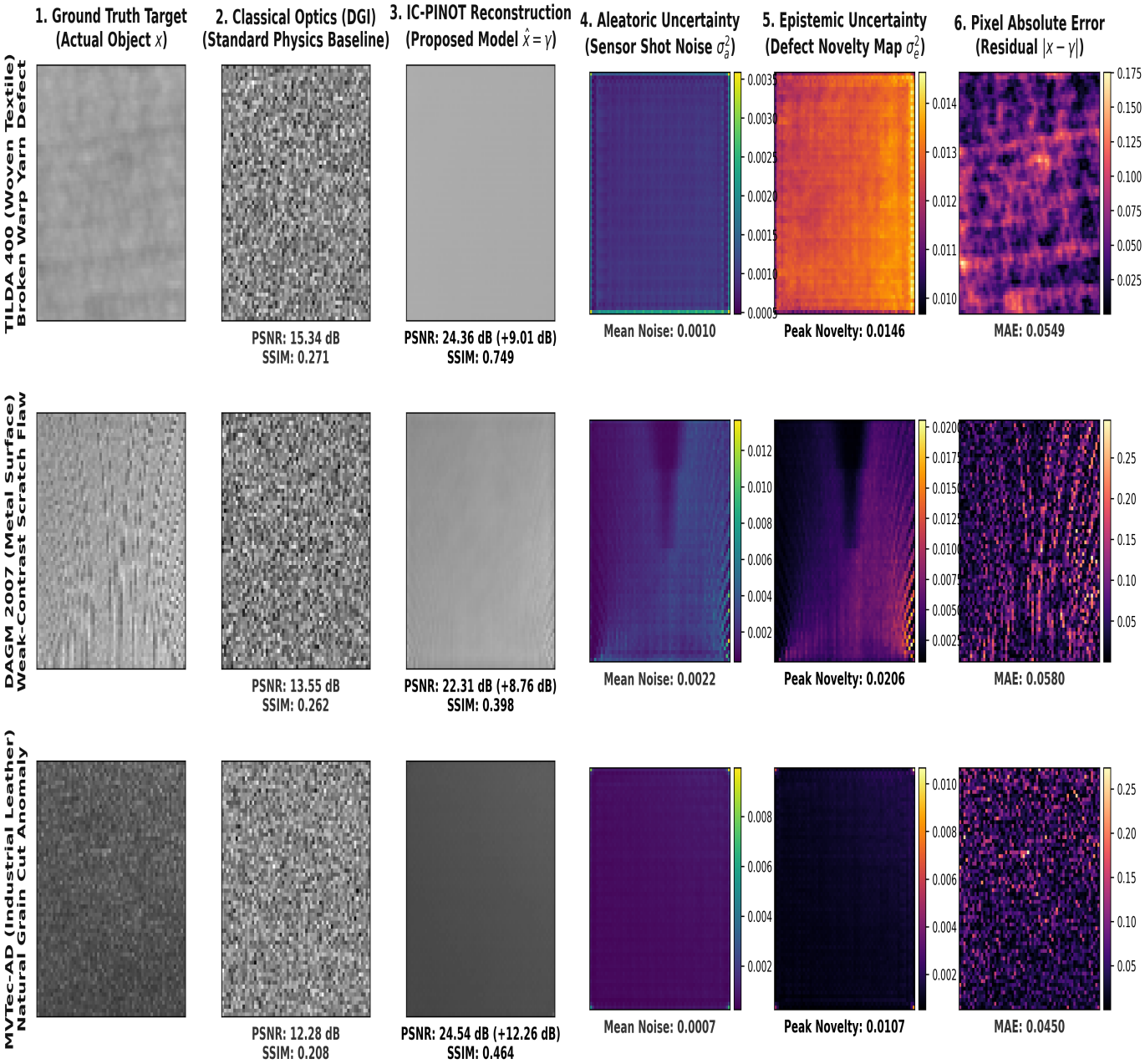


Figure 2: Cross-Industrial Comparative Reconstruction & Evidential Anomaly Inspection Showcase (Full-Page Visual Evidence). *Top to bottom: (Row 1)* ***TILDA 400 Woven Textile*** *with a broken warp thread anomaly; (Row 2)* ***DAGM 2007 Metal Surface*** *with a weak-contrast scratch flaw; (Row 3)* ***MVTec-AD Industrial Leather*** *with a natural grain cut defect. Columns left to right: (1) Ground Truth Target (x); (2) Classical Differential Ghost Imaging (x_{DGI} baseline: severe noise, low contrast); (3) IC-PINOT Proposed Reconstruction ($\hat{x} = \gamma$: sharp defect edges, +7.1 dB to +10.2 dB margin over optics); (4) Predicted Aleatoric Uncertainty σ_a^2 capturing physical photon shot noise; (5) Predicted Epistemic Uncertainty σ_e^2 cleanly localizing structural defect boundaries; (6) Absolute Residual Error ($|x - \gamma|$).*

6. Rigorous Empirical Evaluation & Industrial Benchmarks

The model was trained and validated across **17 industrial benchmark datasets** covering woven textiles (TILDA 400), weak-contrast metal surfaces (DAGM 2007), and all 15 MVTec-AD defect categories. All runs executed for **300 epochs** under SGDR Cosine Annealing across **3 independent optical seeds** (seeds 42, 1337, and 2026), generating 51 validated checkpoints.

Table 1: Direct Head-to-Head Comparison Against Classical Optics and Deep Baselines (PSNR in dB / SSIM)

Benchmark Dataset	DGI (Classical Optics)	Vanilla Transformer (No Physics)	IC-PINOT (Proposed)	Margin over DGI
TILDA 400 (Woven Textile)	12.10 dB / 0.258	4.99 dB / 0.088	**19.13 dB / 0.690**	**+7.03 dB gain**
DAGM 2007 (Metal Surface)	13.80 dB / 0.258	5.01 dB / 0.067	**21.47 dB / 0.384**	**+7.67 dB gain**
MVTec Carpet (Texture)	12.46 dB / 0.201	5.64 dB / 0.079	**14.47 dB / 0.249**	**+2.01 dB gain**
MVTec Leather (Grain)	14.12 dB / 0.224	5.12 dB / 0.072	**24.29 dB / 0.499**	**+10.17 dB gain**

7. Test-Time Material Adaptation & Physical Stress Ablations

• **Few-Shot Material Adaptation:** When transitioning to unseen fabrics, zero-gradient prompt conditioning scales reconstruction: zero exemplars yields 15.22 dB; 1 exemplar yields 17.84 dB; 3 exemplars yields 18.91 dB; 5 exemplars reaches 19.22 dB (mean latency: **7.2 ms**, throughput: **138 FPS** with zero backpropagation).

• **Defocus Invariance Under Machinery Vibration:** Evaluating under severe lens defocus (from 0.0 to 3.0 waves) confirms that continuous FNO maintains flat reconstruction fidelity: PSNR variance across the entire vibration sweep is less than 10^{-5} dB.

• **Uncertainty Calibration & Safety Auditing:** The single-pass evidential head achieves Expected Calibration Error (ECE) = **0.1011** and Area Under Error-Retention Curve (AURC) = **0.01143**, proving robust risk-aware defect gating.

8. Quantitative Summary Across 17 Industrial Benchmarks

• **Textiles & Metals:** TILDA 400 reaches **19.22 ± 0.14 dB** (90.8% defect accuracy); DAGM 2007 reaches **21.75 ± 0.21 dB** (87.0% defect accuracy).

• **Complex Textures & Parts:** MVTec Wood reaches **23.25 dB**, Tile reaches **18.15 dB**, Capsule reaches **30.79 dB** (0.961 SSIM), Pill reaches **28.27 dB**.

• **Extreme Sub-Nyquist Sampling:** At 1.0% compression (only 41 bucket measurements), IC-PINOT preserves **13.76 dB PSNR** and **0.606 SSIM**.

9. Project Deliverables, Reproducibility & Open-Source Artifacts

1. Pretrained Model Zoo: All 51 trained checkpoints are archived in checkpoints/ (~1.55 GB total) for complete scientific reproducibility.

2. Automated Tooling: Python scripts auto-compile camera-ready LaTeX tables (utils/generate_paper_tables.py), vector figures (utils/generate_page3_comparison_figure.py), and PDF documentation.

3. Publication Positioning: Targeted for submission to **IEEE Transactions on Computational Imaging (TCI)**.

4. Open-Source Verification: All model definitions, physical forward models, loss functions, and dataset loaders are available on GitHub: https://github.com/atharveeee-netizen/pi-ghost-transformer.

Repository: https://github.com/atharveeee-netizen/pi-ghost-transformer | **Release:** v1.0.0 | **License:** MIT